

AI ACADEMIC MENTOR

Course	CSE 4104 — Software Development III
Team Name	CSE4104-7C-T05
Section	7C
Project Title	AI Academic Mentor — Personalized Learning & Study Assistant
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Team Information

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User Flow Overview

This document presents the complete user flow for the AI Academic Mentor web application. The flow describes how different user types (Students and Admins) navigate the system — from first visit through authentication, core feature usage, analytics review, and administrative management.

The flow is designed to minimize friction and surface the most valuable AI-powered features within the fewest clicks from login. Every primary action is reachable within two steps from the Dashboard.

User Types

Student (Default Role)	Access to all learning features: AI Chat, Quiz Generator, Note Summarizer, Study Planner, Progress Analytics, Notifications, Settings
Admin (Elevated Role)	All student access plus the Admin Panel: user management, announcements, system analytics, and platform configuration
Guest (Unauthenticated)	Access to Landing Page only; redirected to /login for all protected routes

Detailed Flow Table — Student

Step	Screen	User Action / System Response	Access Level
1	Landing Page (/)	User visits the application at the root URL; sees landing page with feature overview and CTA buttons	Public (no auth required)
2	Register (/register)	New user fills registration form: First Name, Last Name, Email, Student ID, Department, Password, Role (Student/Admin)	Public (no auth required)
3	Login (/login)	Returning user authenticates via email + password; system issues JWT token on success	Public (no auth required)
4	Dashboard (/dashboard)	Post-login home: user views academic metrics (quizzes, streak, score), quick-access cards, and upcoming tasks	Authenticated (Student / Admin)
5	Feature Access	User selects a feature from the sidebar: AI Chat, Quiz Generator, Note Summarizer, or Study Planner	Authenticated (Student)
6	AI Interaction	User interacts with the selected AI-powered feature; Gemini API processes the request and returns a response	Authenticated (Student)
7	Results Display	AI result rendered: chat response, quiz score card, structured summary, or study plan — all saved to MongoDB	Authenticated (Student)
8	Progress Review (/progress)	User reviews performance analytics: topic mastery bars, streak calendar, score trend chart, weak area flags	Authenticated (Student)
9	Notifications (/notifications)	User reviews academic reminders, quiz result alerts, and system messages in the notification center	Authenticated (Student)
10	Settings (/settings)	Optional: user updates profile, theme preference, notification toggles, or language setting	Authenticated (Student)
11	Admin Panel (/admin)	Admin-only: manage users (promote/ban/delete), post announcements, view system analytics, configure platform	Authenticated (Admin role only)

Admin User Flow

The admin flow is a restricted path accessible only to accounts with the 'admin' role. The role is assigned at registration (Role Toggle: Student / Admin) and verified server-side via JWT payload on every request to /admin routes. Students who attempt to access /admin directly receive a 403 Forbidden response.

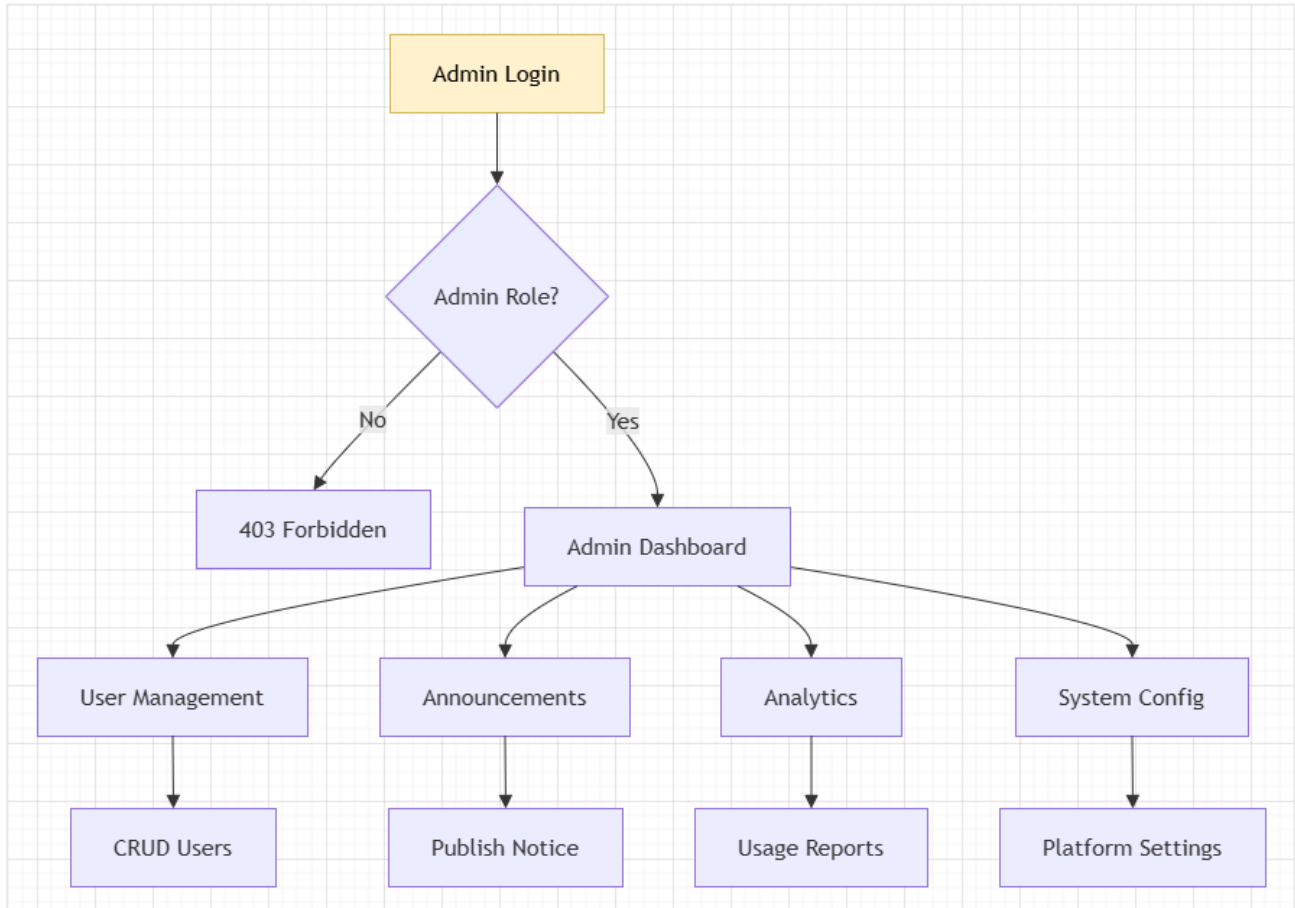


Figure: Admin User Flow

Authentication & Session Flow

Login Flow

- User submits email + password via /login form
- Backend validates credentials against bcrypt-hashed password in MongoDB
- On success: JWT token issued (contains user ID, role, expiry)
- Token stored client-side; attached to all subsequent API requests via Authorization header
- User redirected to /dashboard
- On failure: inline error message displayed — 'Invalid email or password'

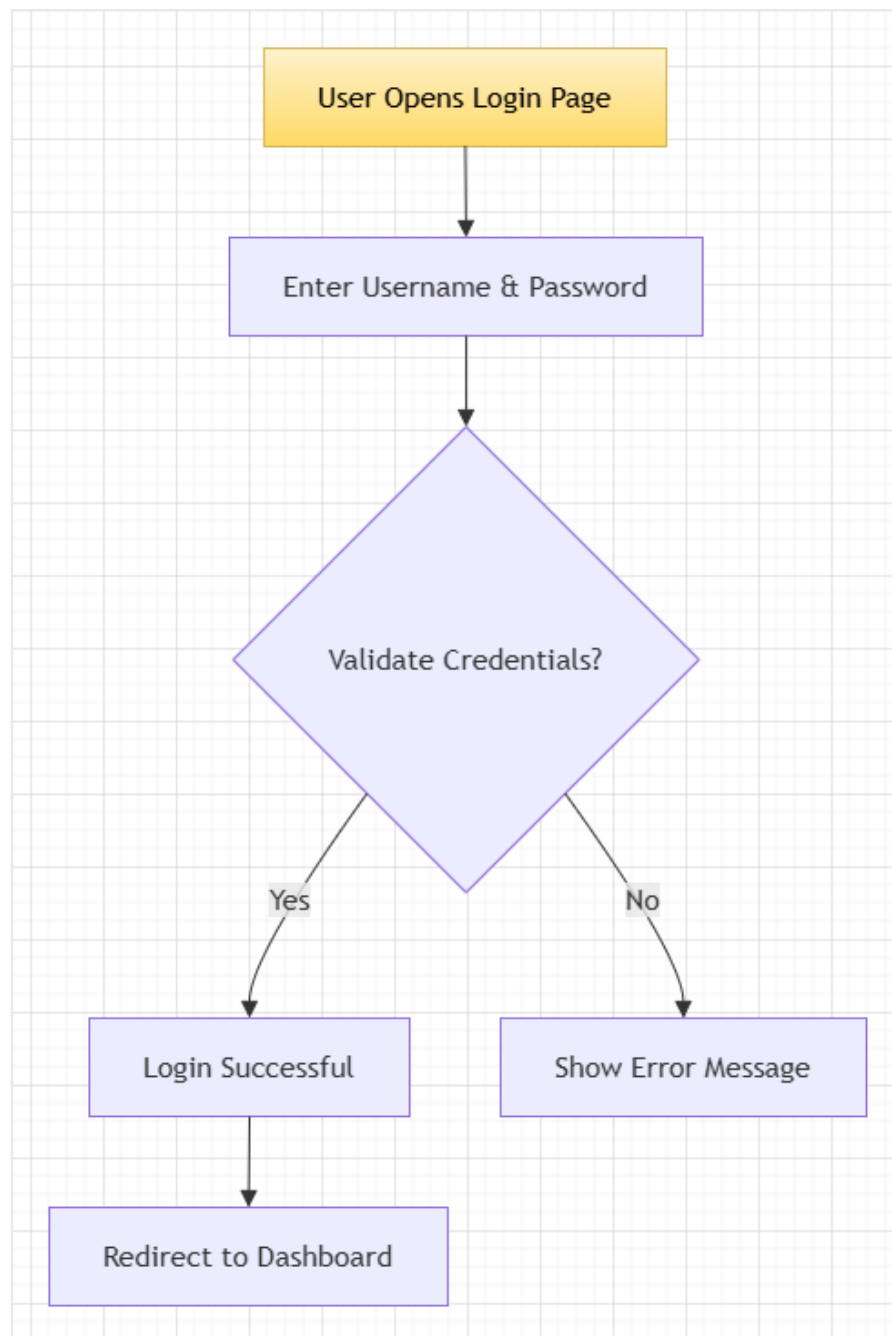


Figure: Login Flow

Session Management

- Protected routes verify JWT on every request; expired or invalid tokens redirect to /login
- Admin routes additionally verify the role field in the JWT payload
- Logout clears the token client-side and redirects to /login

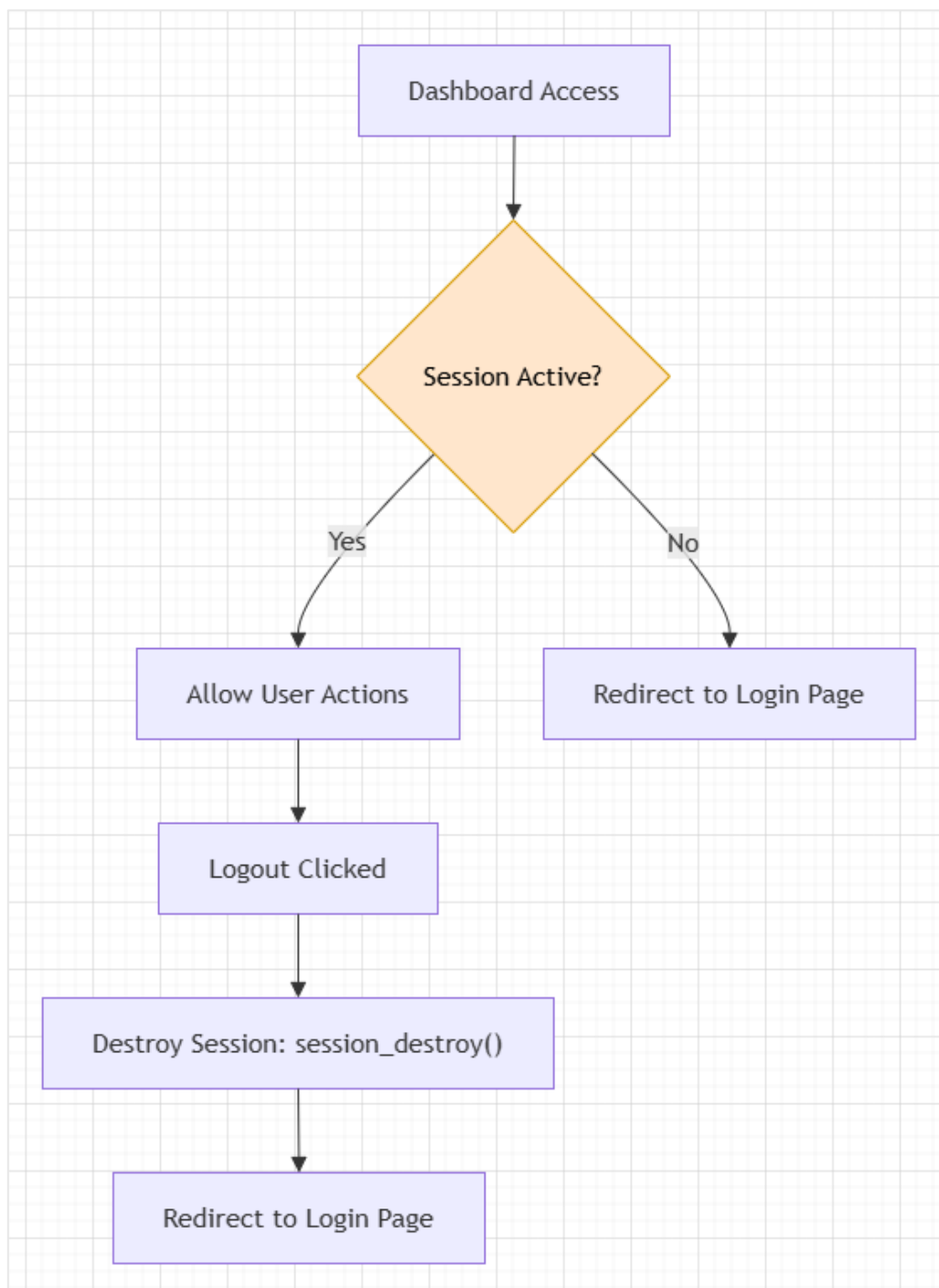


Figure: Session Management

Registration Flow

- User fills: First Name, Last Name, Email, Student ID, Department (CSE/EEE/BBA), Password, Role (Student/Admin)
- Frontend validates required fields and email format before submission
- Backend checks email uniqueness in MongoDB — returns error if duplicate
- Password is hashed with bcrypt (salt rounds: 10) before storage
- On success: user redirected to /login with success message

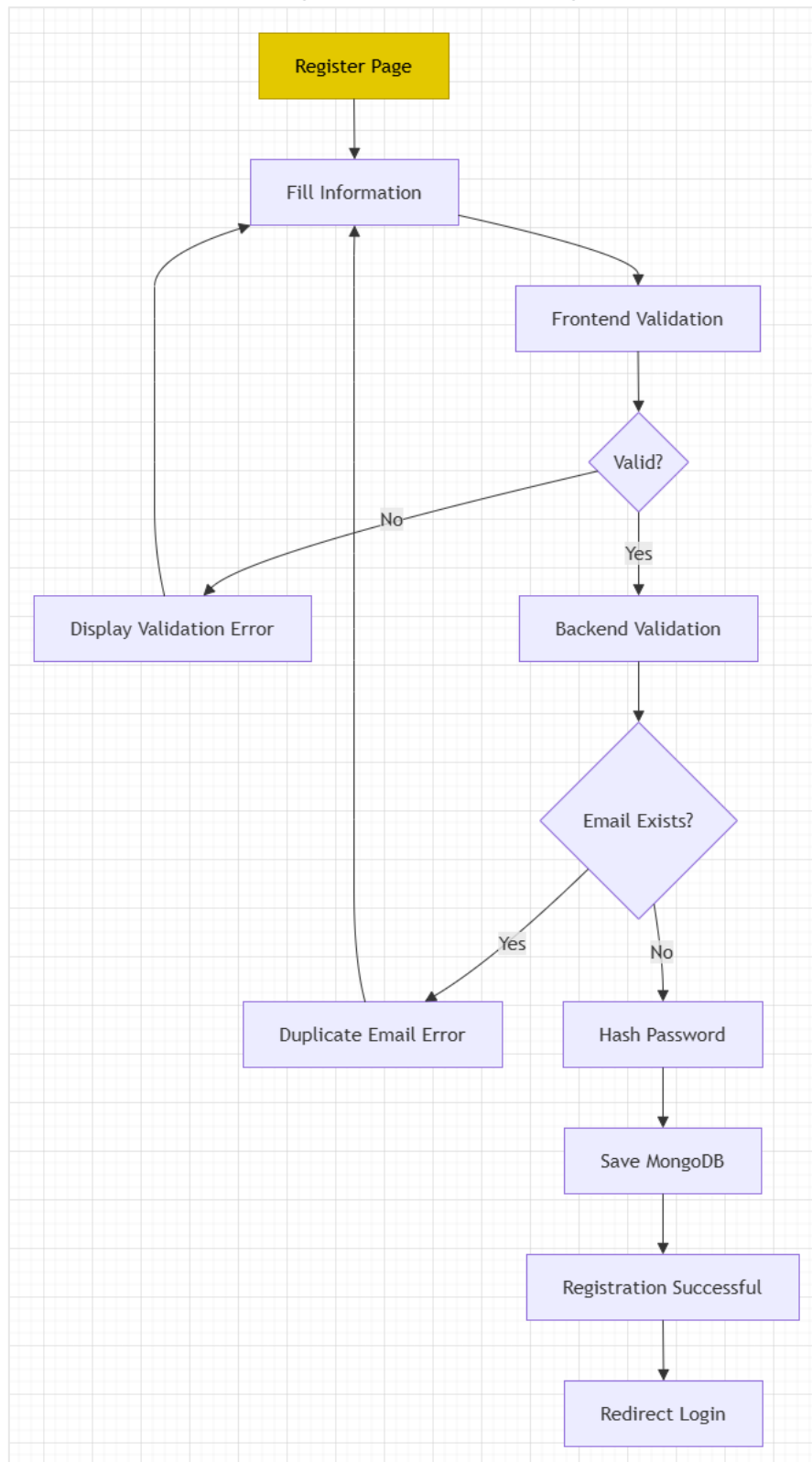


Figure: Registration Flow

Feature-Level Interaction Flows

Dashboard Flow

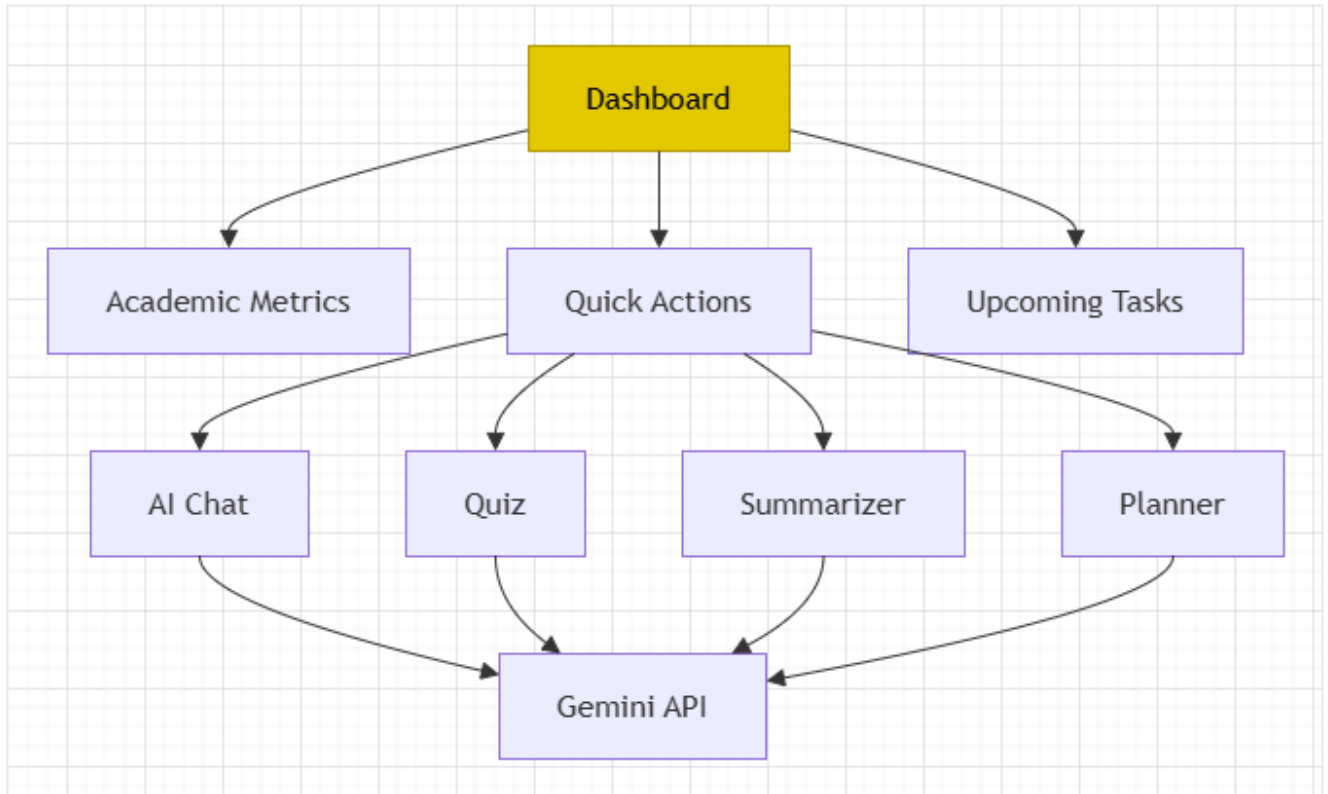


Figure: Dashboard Flow

Notification Flow

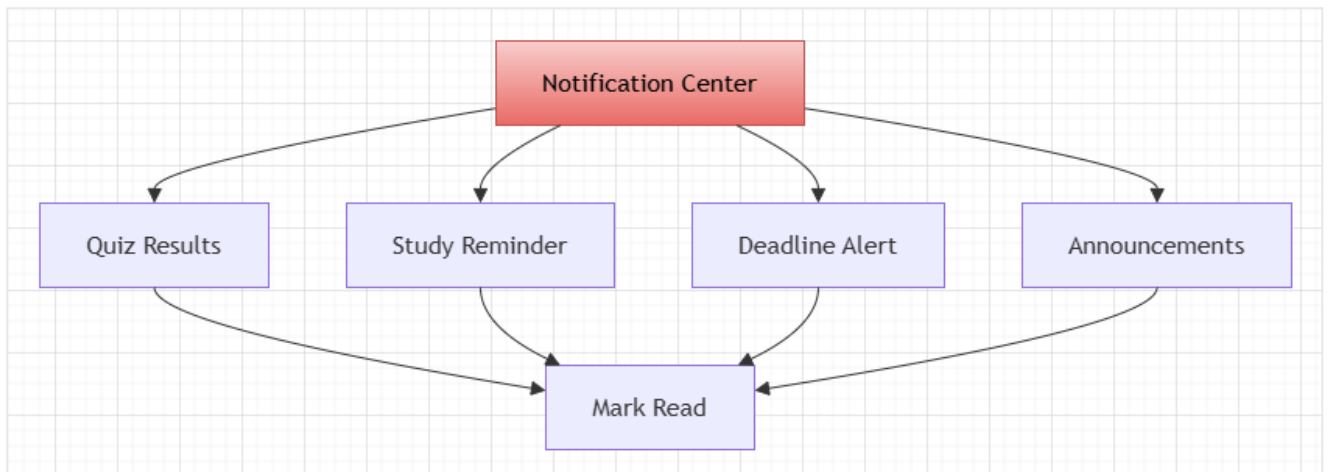


Figure: Notification Flow

Progress Analytics Flow

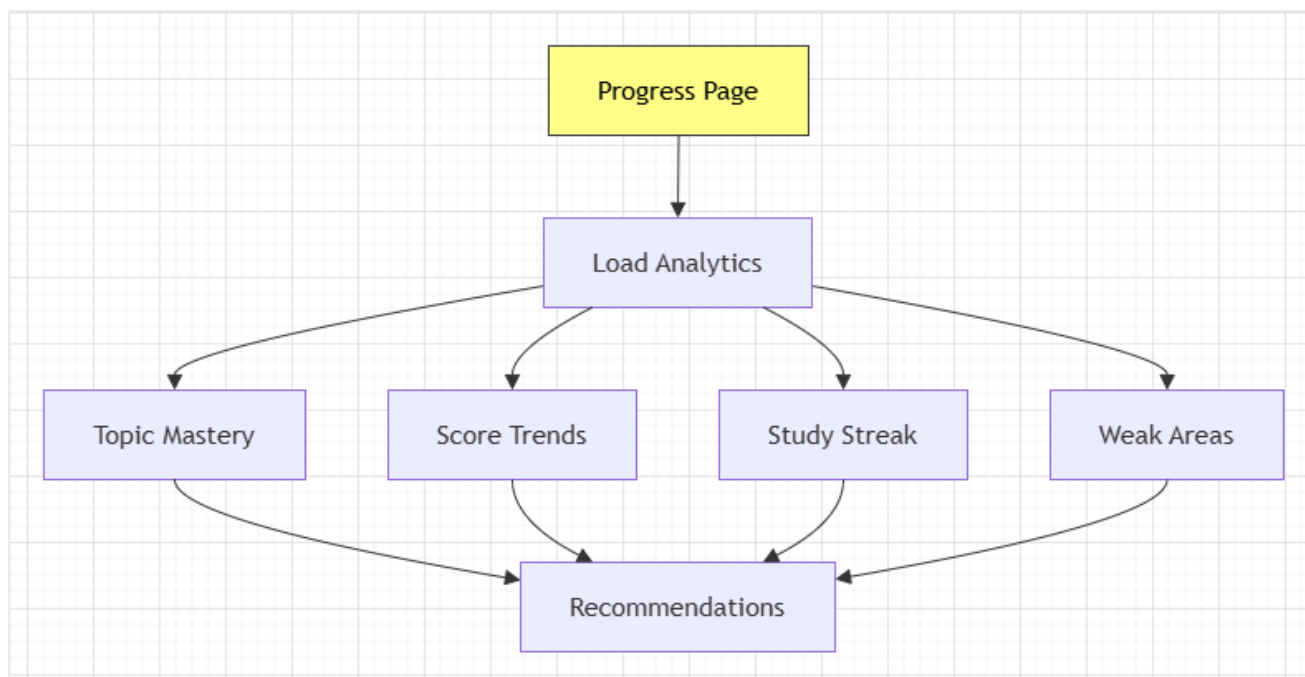


Figure: Progress Analytics Flow

Settings Flow

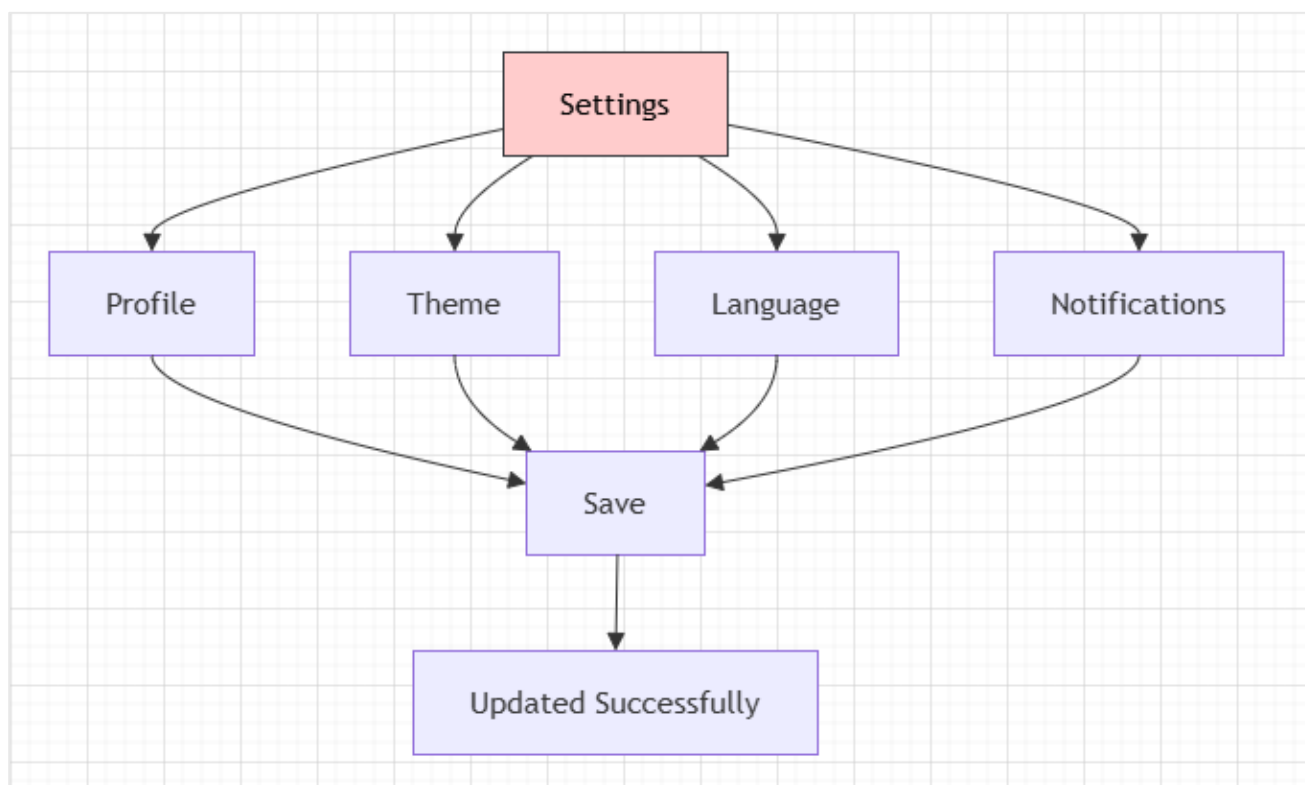


Figure: Settings Flow

AI Chatbot Tutor Flow (/chatbot)

- User selects subject from top dropdown (switches AI context)
- User types question or taps a Suggested Prompt chip
- Request sent to backend proxy → Gemini API
- AI response streamed back and rendered in the message thread
- Conversation history saved to MongoDB session
- User continues conversation or starts a new session

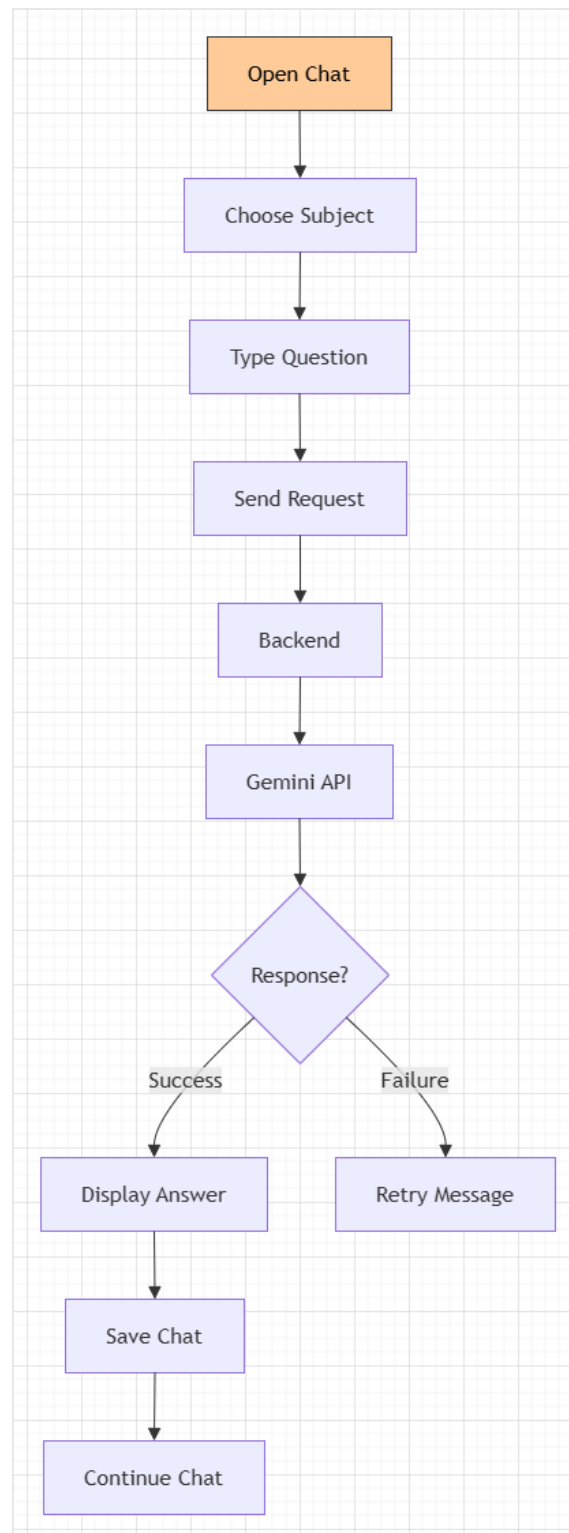


Figure: Chatbot Flow

Quiz Generator Flow (/quiz)

- User configures quiz: Subject, Difficulty (Easy/Medium/Hard), Question Count, Format (MCQ/Short Answer)
- User clicks 'Generate Quiz' — Gemini API generates questions
- Question cards displayed one at a time with progress bar and timer
- User selects answer (MCQ) or types response (Short Answer) for each question
- After final question: Score Summary card shown (percentage ring, correct/wrong counts)
- Score and per-question results saved to MongoDB for analytics
- User can retry, review wrong answers, or return to dashboard

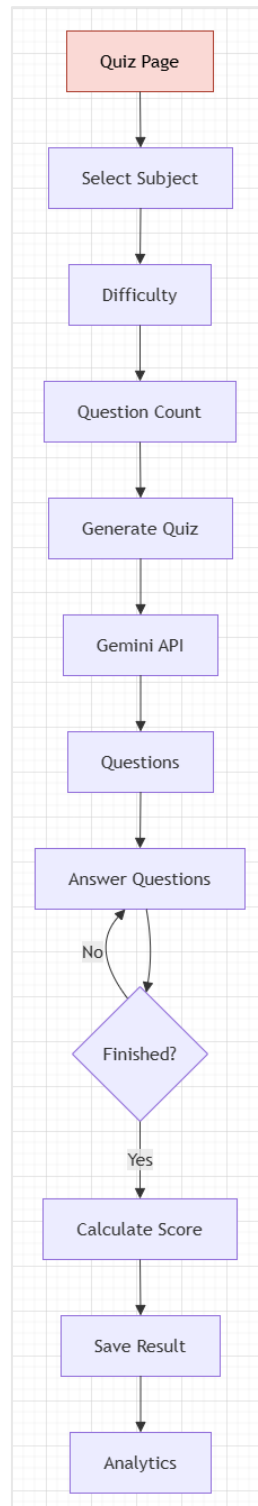


Figure: Quiz Generator Flow

Note Summarizer Flow (/summarizer)

- User pastes lecture notes into textarea OR drags and drops a PDF/text file into the upload zone
- User selects output format: Concise / Detailed / Bullet Points
- User clicks 'Summarize' — Gemini API generates structured summary
- Output card rendered on the right panel with auto-detected subject tag
- User can copy output or download as a text file

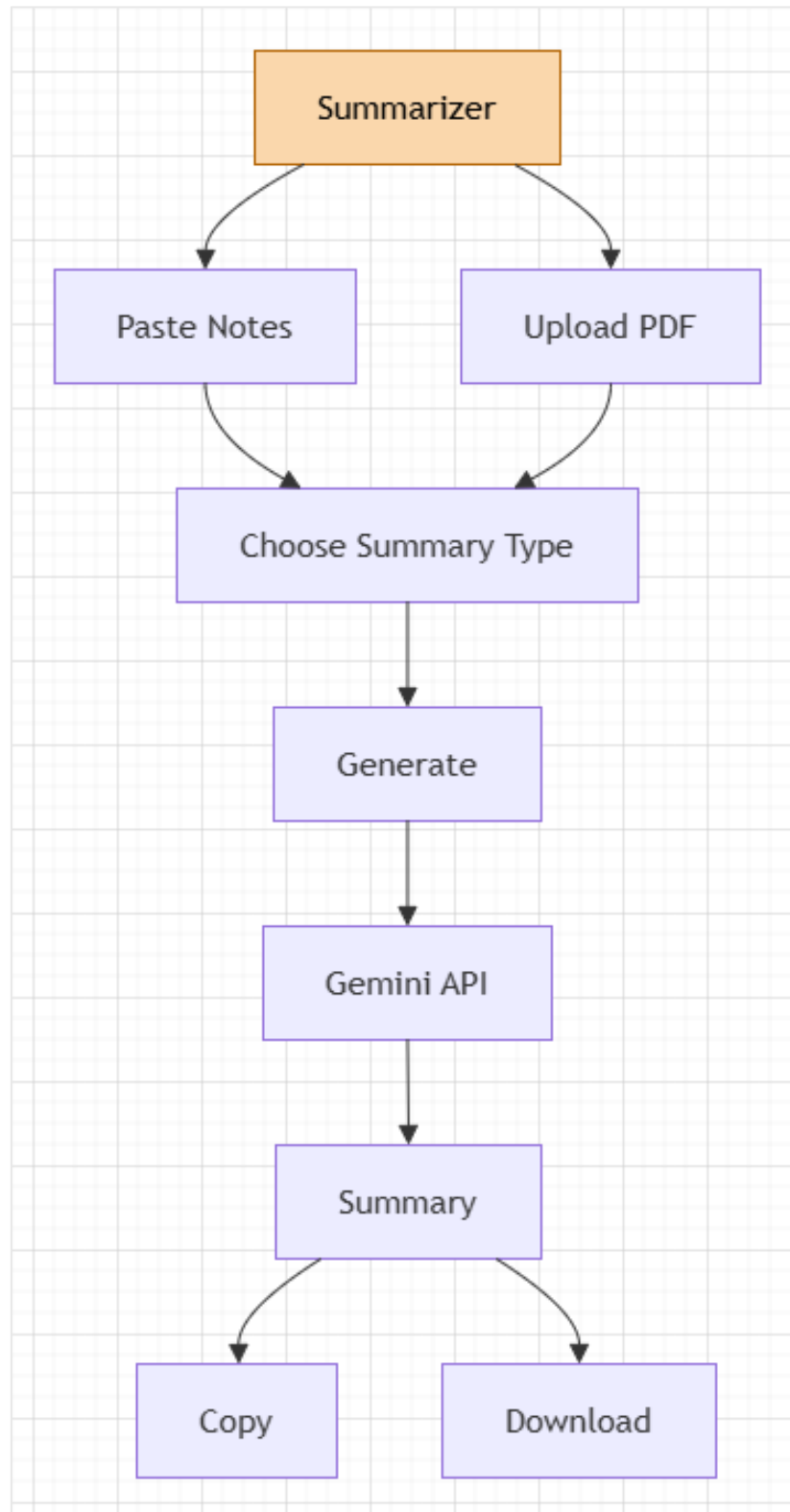


Figure: Note Summarizer Flow

Study Planner Flow (/planner)

- User views the 7-column weekly grid (Mon-Sun) with existing tasks
- User clicks 'Add Task' to open the inline modal form
- User fills: Subject, Date, Time, Duration, Priority (HIGH/MED/LOW)
- Task card appears in the weekly grid, color-coded by subject
- User marks tasks complete — per-day completion percentage updates
- Tasks due within 24 hours trigger a red-bordered Deadline Alert card
- All tasks persisted to MongoDB and synced across sessions

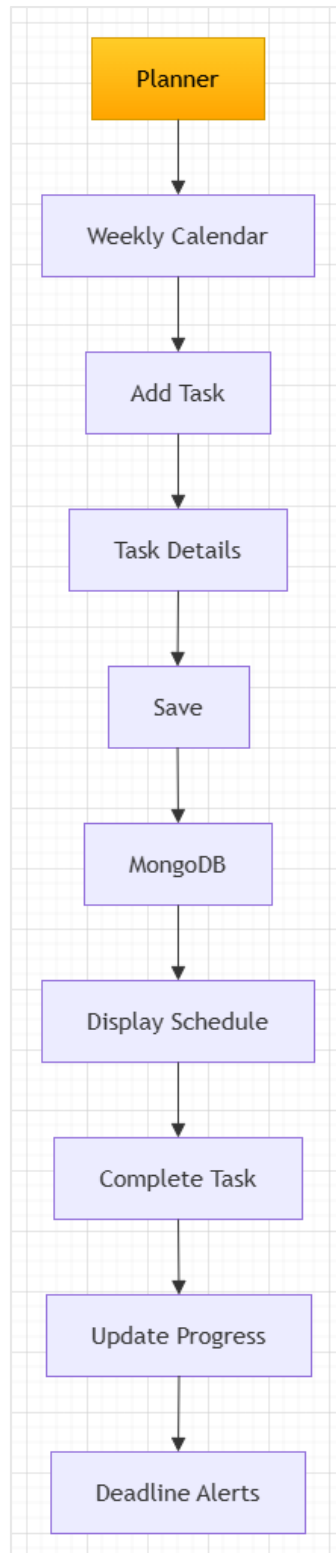


Figure: Study Planner Flow